Machine learning algorithms, mapped.

Sixteen algorithms, from linear regression to diffusion models, each drawn as the idea behind it and classified by how it learns, what it does, and where it is used.

Learns from

Supervised labeled examples

Unsupervised unlabeled data

Task

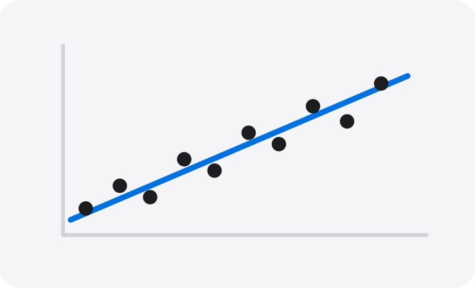
Classification · Clustering · Generation

Used in

▦ Tabular data · ◉ Computer vision · 💬 Natural language · ✦ Generative AI

Classical supervised learning.

Learns from examples that already carry the right answer. The output is a number (regression) or a category (classification).



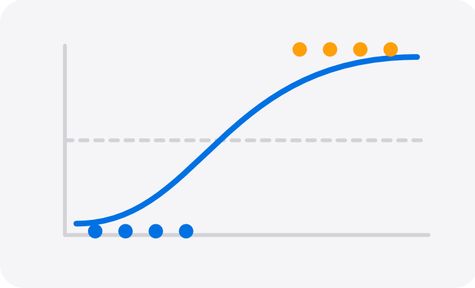
Linear regression

Supervised · Regression

▦ Tabular data

Fits the straight line that minimizes the squared gap to every point.

Price forecasting · Delivery time estimates · Demand projection



Logistic regression

Supervised · Classification

▦ Tabular data

💬 Natural language

Squashes a weighted sum through an S-curve to get a probability, then thresholds it.

Credit scoring · Spam filtering · Churn prediction

Naive Bayes

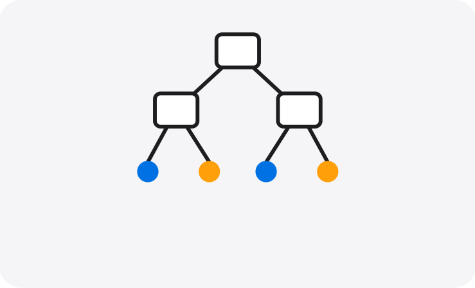
Supervised · Classification

💬 Natural language

▦ Tabular data

Multiplies how likely each feature is under each class and picks the most probable class.

Spam and phishing filters · Email routing · Sentiment baselines



Decision tree

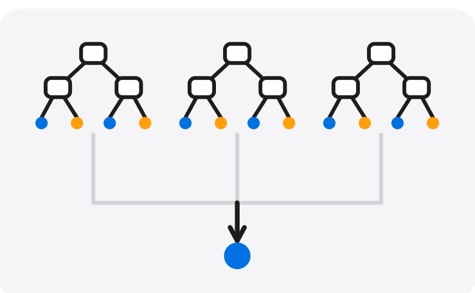
Supervised

Classification, regression

▦ Tabular data

Asks yes-or-no questions one feature at a time, each chosen to make the groups purest.

Readable loan rules · Medical triage · Produce sorting



Random forest

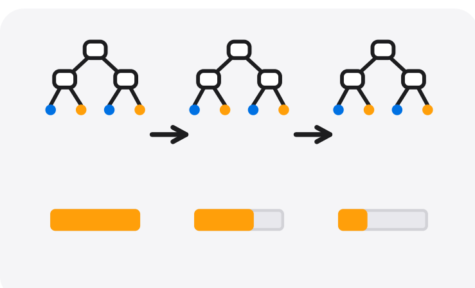
Supervised

Classification, regression

▦ Tabular data

Hundreds of trees grown on random samples of rows and features vote on the answer.

Fraud detection · Churn prediction · Feature importance



Gradient boosting (XGBoost, LightGBM)

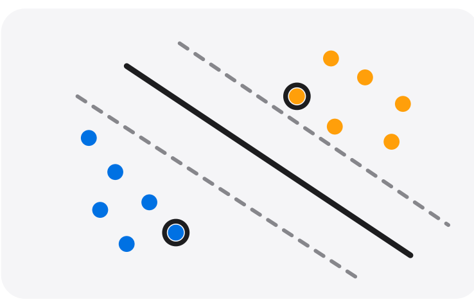
Supervised

Classification, regression

▦ Tabular data

Small trees built in sequence, each one fitted to the errors the earlier ones left.

Click prediction · Risk pricing · Tabular competitions



Support vector machine

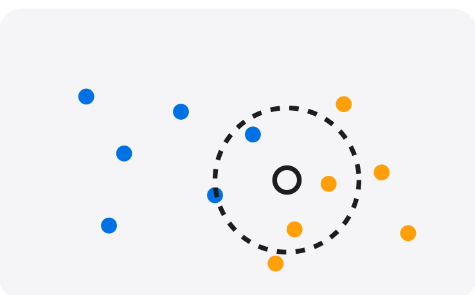
Supervised · Classification

▦ Tabular data

💬 Natural language

Finds the boundary with the widest margin between classes; kernels let it curve.

Text categorization · Digit recognition · Bioinformatics



K-nearest neighbors

Supervised

Classification, regression

▦ Tabular data

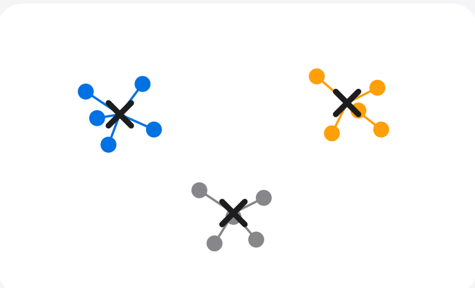
◉ Computer vision

Finds the k closest stored examples and takes their majority label.

Similar-item recommendations · Anomaly detection · Small image baselines

Classical unsupervised learning.

Works on unlabeled data and finds structure in it: groups, directions of variation, outliers.



K-means clustering

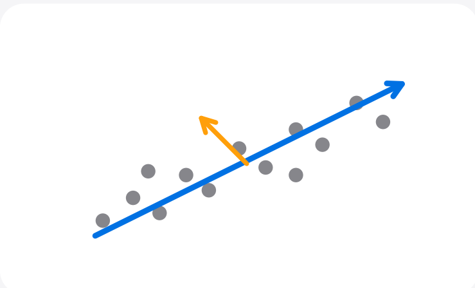
Unsupervised · Clustering

▦ Tabular data

◉ Computer vision

Assigns points to the nearest of k centers, moves each center to the mean, repeats.

Customer segmentation · Color quantization · Ticket grouping



Principal component analysis

Unsupervised

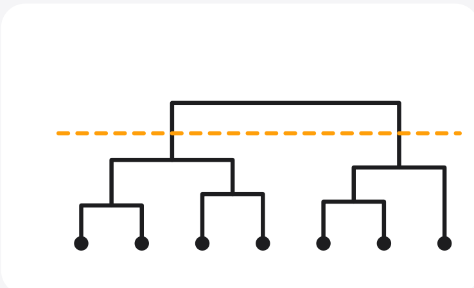
Dimensionality reduction

▦ Tabular data

◉ Computer vision

Projects the data onto the few directions along which it varies most.

Feature compression · 2D visualization · Noise reduction



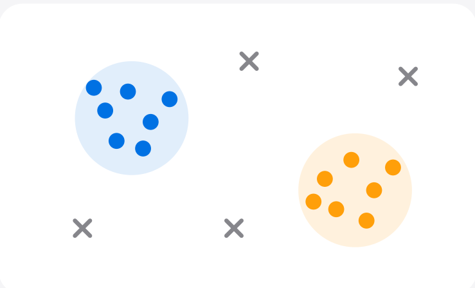
Hierarchical clustering

Unsupervised · Clustering

▦ Tabular data

Merges the two closest clusters again and again, building a tree you can cut at any height.

Customer taxonomies · Gene expression groups · Nested topics



DBSCAN

Unsupervised

Clustering, anomaly detection

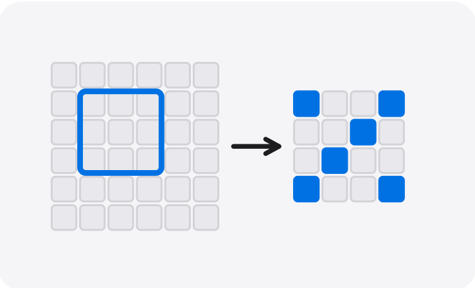
▦ Tabular data

Grows clusters out of dense regions and leaves sparse points as noise.

Transaction anomalies · Geographic hotspots · Unknown cluster counts

Deep learning.

Neural networks with many layers that learn their own features from raw pixels, tokens, or audio. These power computer vision, language models, and generative AI.



Convolutional neural network

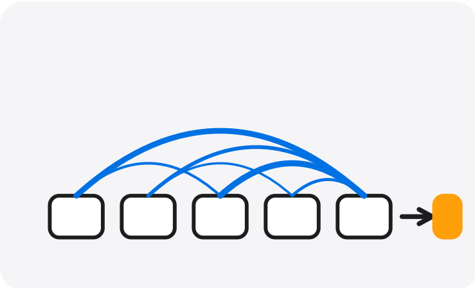
Supervised

Classification, detection

◉ Computer vision

Slides learned filters across the image, building edges into textures into objects.

Image classification · Object detection · Medical imaging



Transformer (BERT, GPT)

Unsupervised

Language modeling, generation

self-supervised pretraining, then supervised fine-tuning

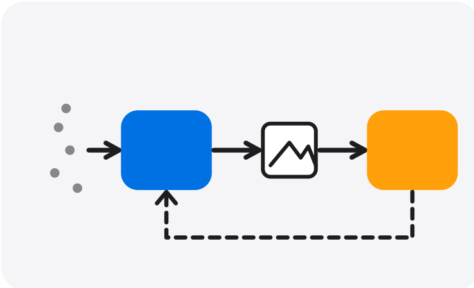
💬 Natural language

✦ Generative AI

◉ Computer vision

Attention lets every token weigh every other token, in parallel, to predict what comes next.

Chat assistants · Translation and search · RAG question answering



Generative adversarial network

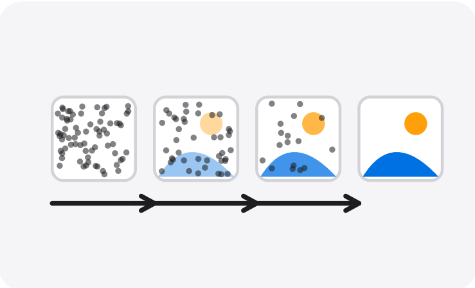
Unsupervised · Generation

✦ Generative AI

◉ Computer vision

A generator forges samples, a discriminator calls them out, and both improve until the fakes pass.

Face synthesis · Sketch to photo · Data augmentation



Diffusion model (Stable Diffusion, DALL-E)

Unsupervised · Generation

self-supervised

✦ Generative AI

◉ Computer vision

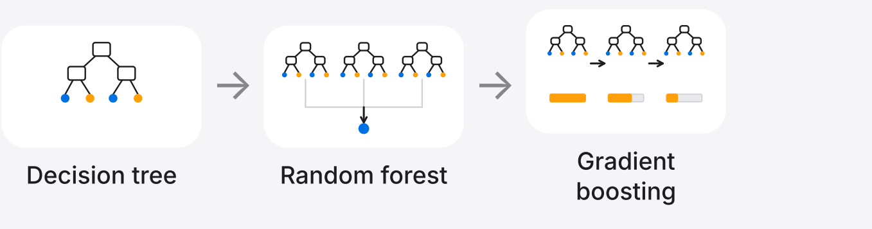
Learns to remove noise step by step, so it can turn pure noise into a new image guided by a prompt.

Text-to-image · Inpainting · Video and audio generation

How they connect.

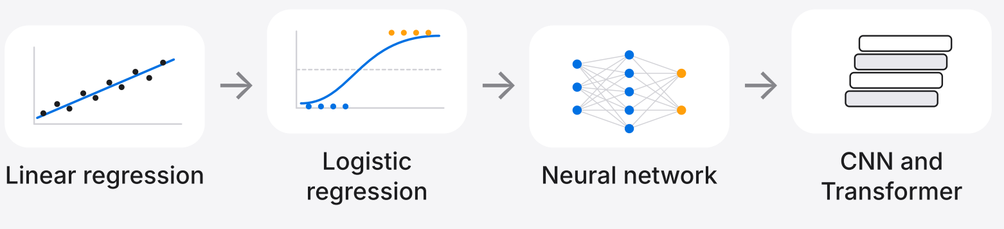
The bands sort algorithms by how they learn. These five chains show how they descend from one another, which is usually the fastest way to choose between neighbors on the map.

Trees grow into ensembles



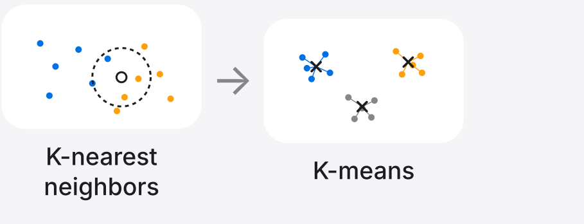
A single tree is easy to read but overfits. A random forest averages hundreds of trees grown on random samples; gradient boosting grows them in sequence, each one correcting the errors of the last. Accuracy rises at each step and interpretability falls.

From a line to a network



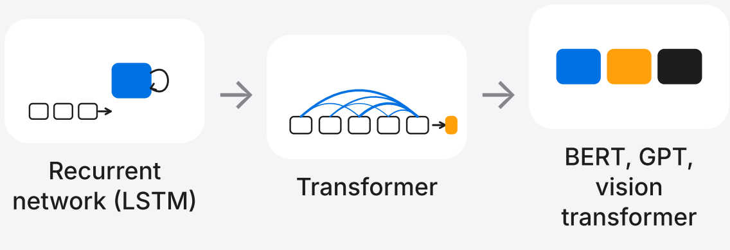
Logistic regression is linear regression with an S-shaped curve on the output. A neural network stacks many such units in layers. Convolutional networks and transformers are neural networks with structure added for images and sequences.

Same distance, different question



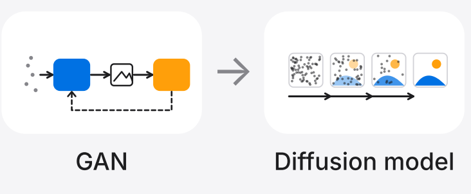
Both measure distance between points, so both need scaled features. K-nearest neighbors uses the labels of nearby points (supervised); K-means has no labels and discovers the groups itself (unsupervised).

Attention replaced recurrence



Recurrent networks read text one token at a time and struggled with long passages. The transformer reads the whole sequence at once through attention, which made large language models possible and then carried over to images.

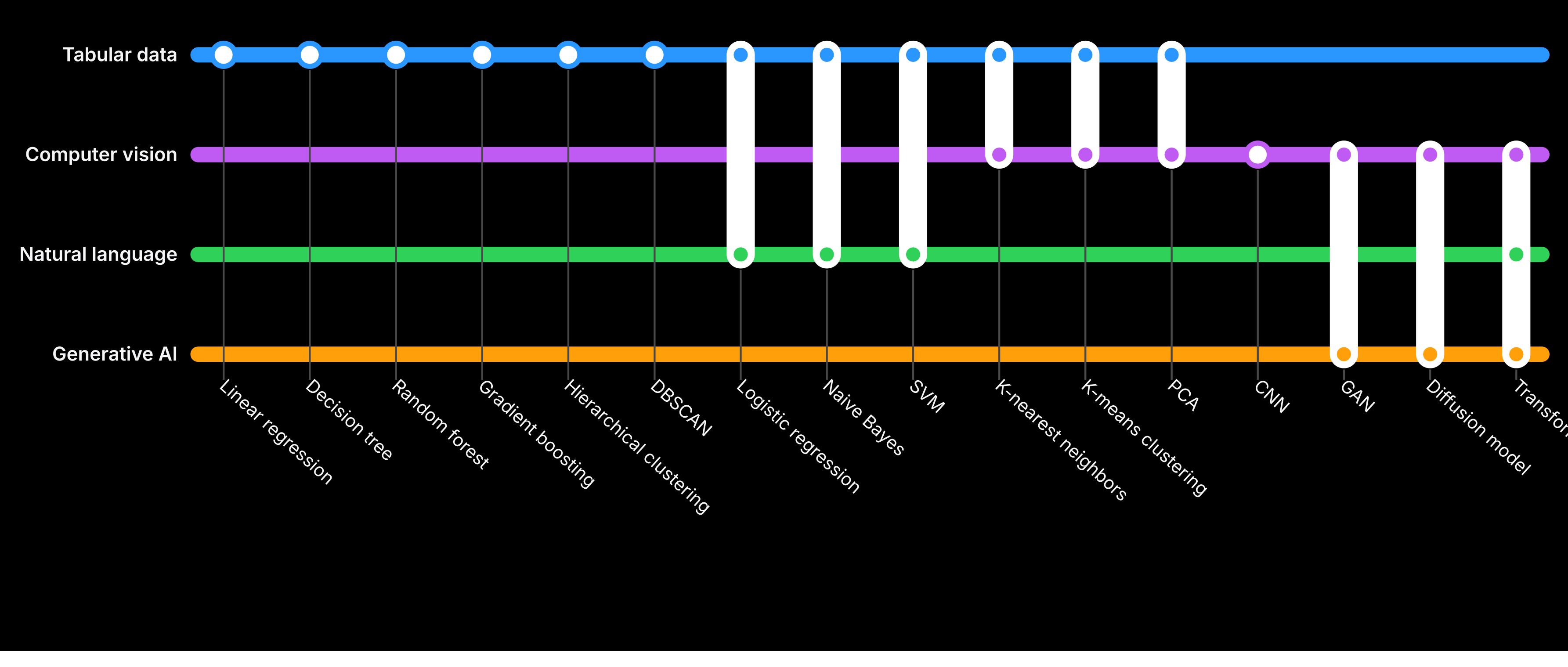
Two roads to generation



A GAN learns through a contest between two networks: fast to sample from, hard to train stably. A diffusion model learns to reverse added noise: slower to sample, stable to train, higher quality. Diffusion has largely replaced GANs for image generation since 2022.

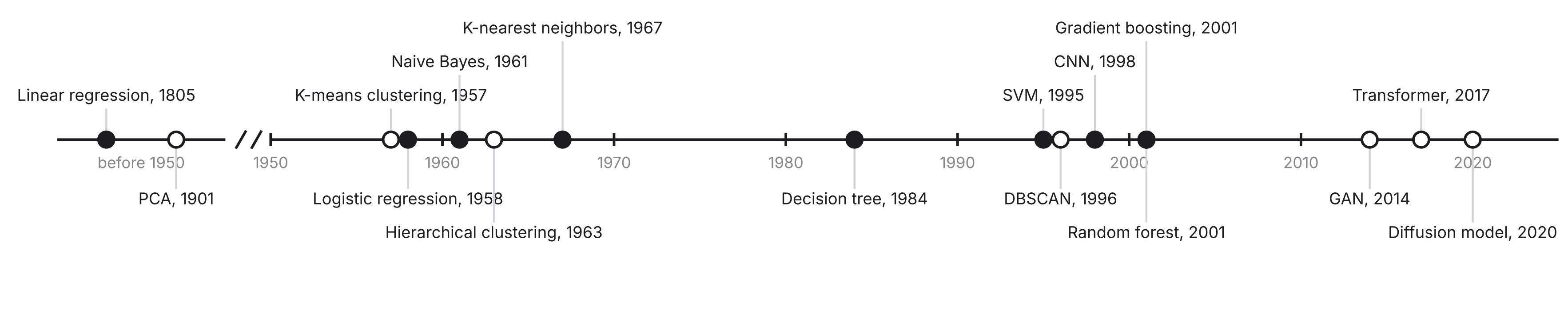
Where the lines meet.

Four domains as four lines. Each algorithm is a station; where a station spans several lines, the same algorithm serves several domains. Tabular data runs almost entirely on classical methods; vision, language, and generative AI are served by deep learning, and the transformer is the only station on three lines.



Seventy years, one line.

When each algorithm was introduced. Filled dots are supervised, hollow dots unsupervised. The classical toolkit was complete by about 2001; everything that serves vision, language, and generation at scale arrived after 2012, when deep learning became practical on GPUs.



Transformers, GANs, and diffusion models are marked unsupervised because their training signal comes from the data itself (the next token, the real-or-fake contest, the noise that was added), not from human labels. A supervised fine-tuning stage usually follows for a specific task.

Domain chips show where each algorithm is a standard choice today; most can be pushed into other domains with suitable feature engineering.

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November 2026